Harrison Crosse

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EMPLOYMENT

Docker

Senior Software Engineer 2025 - Present

- Co-designed Marlin, Docker's event platform, to handle tens of thousands of events per second. Established its SDK and ingestion architecture and guided expansion into stream processing and audit-log analytics.
- Built Marlin SDKs for browser, Node.js, and Rust around shared protobuf contracts. Productionized them with batching, retries, integration tests, and release automation.
- Designed Marlin's privacy architecture and annotation-driven PII handling across SDKs, ingestion workers, and Iceberg, supporting Docker's ISO 27701 certification.

Calendly

Senior Data Engineer 2024 - 2025

Data Engineer 2022 - 2024

- Built Calendly's data platform from the ground up, migrating Postgres and event data from Redshift to BigQuery and establishing Airflow, CI/CD, data quality, and developer tooling.
- Processed 50 TB of malformed Segment replay data with Spark for BigQuery.
- Built production Flink pipelines processing thousands of records per second with hundreds of gigabytes of state. Added observability, blue-green deployments, data-quality checks for cutovers, and E2E tests.
- Replaced unreliable Google Sheets-defined schemas with a GitOps Data Contracts workflow across hundreds of Segment event types.

Amobee

Software Engineer, Data Systems 2021 - 2022

- Migrated clickstream processing from batch Spark to Structured Streaming, making data continuously available downstream.
- Moved Spark workloads from EMR to Kubernetes with Spark Operator, standardizing deployments and eliminating EMR service charges.

Sayari

Data Engineer 2020 - 2021

- Built scraping and fastText NER pipelines to transform corporate registry data into structured entities and social graphs.
- Migrated Sayari's cloud infrastructure from GCP to AWS GovCloud using Terraform and Docker.

Booz Allen Hamilton

Data Scientist 2019 - 2020

- Modeled Marine Corps unit performance for the Inspector General using clustering and gradient-boosted trees.

EDUCATION

University of Mary Washington

B.S. Computer Science 2019

Minor in Data Science

SELECTED WORK

DataFusion / Ballista

Upstreamed non-blocking distributed shuffle execution by exposing async batch partitioning in DataFusion and offloading Ballista shuffle writes from query workers.

DataFusion #21341 Ballista #1537

Apache Iceberg / Arrow Go

Improved Parquet writer compatibility by adding table-level page-version configuration to Iceberg and fixing invalid root-schema serialization in Arrow Go.

Iceberg #15700 Arrow Go #723

TOOLS

LANGUAGES

Bash C Go Java Python Rust SQL TypeScript

DATA SYSTEMS

Airflow Arrow BigQuery ClickHouse dbt Debezium Flink Iceberg Kafka Parquet PostgreSQL Protobuf Snowflake Spark

INFRASTRUCTURE

Argo AWS Docker GCP GitHub Actions Grafana Helm Kubernetes Linux Terraform